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RESEARCH ARTICLE

Dimethyl Fumarate Delays Multiple Sclerosis in Radiologically Isolated Syndrome

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Objective: The radiologically isolated syndrome (RIS) represents the earliest detectable pre-clinical phase of multiple sclerosis (MS). This study evaluated the impact of therapeutic intervention in preventing first symptom manifestation at this stage in the disease spectrum.

Methods: We conducted a multi-center, randomized, double-blinded, placebo-controlled study involving people with RIS. Individuals without clinical symptoms typical of MS but with incidental brain MRI anomalies consistent with central nervous system (CNS) demyelination were included. Within 12 MS centers in the United States, participants were randomly assigned 1:1 to oral dimethyl fumarate (DMF) 240 mg twice daily or placebo. The primary endpoint was the time to onset of clinical symptoms attributable to a CNS demyelinating event within a follow-up period of 96 weeks. An intention-to-treat analysis was applied to all participating individuals in the primary and safety investigations. The study is registered at ClinicalTrials.gov, NCT02739542 (ARISE).

Results: Participants from 12 centers were recruited from March 9, 2016, to October 31, 2019, with 44 people randomized to dimethyl fumarate and 43 to placebo. Following DMF treatment, the risk of a first clinical demyelinating event during the 96-week study period was highly reduced in the unadjusted Cox proportional-hazards regression model (hazard ratio [HR] = 0.18, 95% confidence interval [CI] = 0.05–0.63, $p = 0.007$). More moderate adverse reactions were present in the DMF (34 [32%]) than placebo groups (19 [21%]) but severe events were similar (DMF, 3 [5%]; placebo, 4 [9%]).

Interpretation: This is the first randomized clinical trial demonstrating the benefit of a disease-modifying therapy in preventing a first acute clinical event in people with RIS.

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Multiple sclerosis (MS) is a treatable, heterogeneous inflammatory demyelinating and degenerative condition of the CNS resulting in neurological impairment in young adults and chronically affecting nearly 1 million people in the United States.[1] The diagnosis requires fulfillment of clinical, radiological, and laboratory criteria meeting dissemination in time and space and with the exclusion of other medical conditions that provide a better explanation for the acquired findings.[2] Following the development of a symptom consistent with inflammatory demyelination, clinical classifications of clinically isolated syndrome (CIS), relapsing–remitting (RR), secondary progressive (SP), or primary progressive (PP) MS may be applied.[3] Intervention with disease-modifying therapies (DMTs) in CIS delays the onset of RRMS.[4–7] However, even treatment at this early symptomatic phase may not be soon enough given the risk for incomplete recovery and permanent disability.[8] Moreover, no current prevention strategies for PPMS exist without preceding clinical symptoms. There is an unmet need to properly identify and effectively treat people as early as possible to delay or prevent the clinical onset of MS.

Over the past decade, with the widespread use of MRI, recognizing individuals in routine clinical practice with incidental brain or spinal cord anomalies highly suggestive of CNS demyelination has increased. As originally described in 2009,[9] these individuals with no definitive symptoms related to inflammatory demyelination are classified as having "radiologically isolated syndrome (RIS)". Prospective cohort studies indicate that a meaningful proportion of people, but not all, are at risk for developing a first acute demyelinating event[10–12] or the onset of a neurological symptom with accumulating disability, thereby fulfilling diagnostic criteria for CIS/RR, or PPMS, respectively.[13] The occurrence of RIS, and risk for developing MS, have also been recognized in children.[14, 15]

The impact of DMTs in preventing the clinical onset of MS in individuals with RIS is unknown. Here, in the first randomized, double-blinded, placebo-controlled trial in a cohort of people with RIS, we tested the impact of DMF (Tecfidera®)[16, 17] in the prevention of clinical symptoms and MRI worsening related to MS. The use of DMF was selected given the FDA approval for use in relapsing forms of MS, existing safety and efficacy data, the oral mode of delivery, and immunomodulatory characteristics.

Methods

Study Design

This multi-center, randomized, double-blind, placebo-controlled, 96-week study was performed within 12 enrolling MS Center of Excellence sites in the United States that was supported by international advisory, safety monitoring, and independent core clinical and MRI committees.

The study was performed in accordance with the International Conference on Harmonization Guidelines for Good Clinical Practice[18] and the Declaration of Helsinki. The Institutional Review Boards approved the protocol at all participating centers.

Participants

Key eligibility criteria included males and females ≥18 years of age meeting 2009 RIS criteria,[9] identified RIS participants with index MRI date ≥2009, incidental anomalies identified on MRI of the brain or spinal cord with the primary reason for the acquired MRI resulting from an evaluation unrelated to MS, and no clinically apparent neurologic impairments that could be accounted for by the MRI anomalies. Incidentally observed MRI anomalies highly suggestive of CNS demyelination were required to fulfill at least 3 of 4 of the following criteria: (1) ≥9 T2-weighted hyperintense lesions or ≥1 gadolinium-enhancing lesion, (2) ≥3 periventricular lesions, (3) ≥1 juxtacortical lesion, and (4) ≥1 infratentorial lesion. CNS anomalies were required to be inconsistent with a vascular pattern and have characteristic features of demyelinating lesions. The observed anomalies were also required to not account for clinically apparent neurological impairments. Key exclusion criteria were pregnancy or nursing, incomplete medical data, history of remitting clinical symptoms compatible with MS lasting >24 hours prior to or at any time after the index RIS MRI study, paroxysmal symptoms associated with MS, CNS MRI anomalies better accounted for by another disease process, participant unwillingness or inability to comply with requirements of the study protocol, exposure to a disease-modifying therapy (DMT) within the past 3 months, exposure to high-dose glucocorticosteroid treatment within the past 30 days, and participation in other clinical trials involving treatment with a DMT.

Written informed consent was obtained at the screening visit from all study participants.

Randomization and Masking

Study participants were randomly assigned (1:1) to receive oral DMF or placebo. The allocation sequence was acquired using web-based interactive response technology with block randomization methods (block size of 4) from Almac Clinical Services (Durham, NC) and concealment was achieved through code break cards. Principal investigators at each site enrolled participants. All study investigators, site personnel, and participants were masked to

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treatment allocation. Medication and placebo capsules along with packaging were exact in appearance and were identical to those used in the pivotal DMF studies.[16, 17]

Procedures

During the 96-week double-blind treatment period, participants received oral DMF 120 mg twice daily for 7 days followed by 240 mg by mouth twice daily, consistent with the FDA-approved labelling until the end of study date, or placebo. The core clinical committee provided independent reviews of prior medical data and confirmation of inclusion criteria for study entry. This committee, unaware of randomization assignments, also adjudicated clinical relapses related to inflammatory demyelination. The MRI committee performed independent reviews of clinical MRI data and confirmed imaging criteria for study inclusion.

Key clinical data, including the Expanded Disability Status Scale (EDSS), Multiple Sclerosis Functional Composite (MSFC), and Patient Reported Outcome Assessments were acquired at weeks 0, 48, and 96. Brain MRI data were acquired at weeks 0 and 96. Blood draws were performed at weeks 0, 12, 24, 48, 72, and 96. At the time of first clinical relapse, early withdrawal, or unscheduled visit, a physical examination, EDSS, MSFC, and blood draw were performed. Lumbar punctures for cerebrospinal fluid analysis were optional and blood samples were acquired at weeks 0, 48, and 96 weeks. Adverse events were captured throughout the duration of the study.

Outcomes

The prespecified primary endpoint was the time to development of a first acute, adjudicated, clinical event or a progressive neurologic symptom suggestive of CNS demyelination from study enrollment. An acute neurological event was defined by a clinical symptom localized to the optic nerve, brainstem, cerebellum, spinal cord, or long sensory or motor tracts, lasting >24 hours and followed by a period of symptom improvement. The onset of a non-acute primary demyelinating event was defined as gradual worsening of a neurologic symptom over at least 12 months, fulfilling criteria for primary progressive MS.[19]

MRI analysis was performed centrally by the Advanced in MS (AIMS) laboratory (D.P., C.J.A.) at the University of Southern California. Prespecified secondary endpoints included the number of new and/or newly-enlarging T2-weighted hyperintense lesions, the number of gadolinium-enhancing lesions, changes in T2-lesion volume, and brain atrophy measures at 96 weeks. A standardized MRI protocol was used across centers and included a non-enhanced 2D T2/PD-weighted dual-echo sequence ($1 \times 1 \times 3$ mm^3), pre- and post-gadolinium-enhanced 3D T1-weighted gradient echo sequences ($1 \times 1 \times 1$ mm^3), and either a 2D ($1 \times 1 \times 3$ mm^3) or 3D FLAIR sequence. MRI scans were performed at months 0 and 96. The AIMS laboratory evaluated screening MRI data to ensure that the appropriate size, shape, number, and distribution pattern of lesions within the CNS for study entry were met. All MRI data were assessed and post-processed without knowledge of treatment assignment. T2-lesion volume was determined using previously described methods.[20]

Study participants who met the primary endpoint or were intolerant to the randomized therapy were withdrawn from the study. Subjects on blinded study treatment with identified intolerance or those who experienced an adjudicated clinical attack were invited for open-label management with DMF or peg-interferon beta-1a (Plegridy®).[21] For those individuals who were intolerant to treatment but did not meet the primary endpoint, continued follow-up assessments and MRI within the study were encouraged.

Safety assessments included adverse events, physical and neurological examinations, and clinical laboratory parameters. Treatment-emergent adverse events were defined as an untoward medical occurrence in a clinical investigation subject administered a pharmaceutical product that occurs after the first dose of study treatment through week 96 or end of study visit or early withdrawal/discontinuation visit, whether or not considered related to the study treatment. Signs and symptoms present at baseline were only considered adverse events if they worsened while the subject was on the study drug. These events were also stratified as mild, moderate, or severe.

Statistical Analysis

Sample size calculations using Cox proportional hazards regression indicated that 80 participants per arm would have been needed to have 80% power to detect a 50% reduction in the risk of the first clinical event, assuming that 25% of participants experience a first clinical event at 96 weeks and a trial duration of 240 weeks. After the study investigators recognized that recruitment was much slower than planned, a protocol amendment was set, including a Bayesian analysis of the data set as a secondary analysis. The Bayesian approach is based on combining a prior distribution for the treatment effect with the results observed within the trial (representing the likelihood), increasing the power of the study compared to more traditional sample size assumptions. The prior distribution for the effect of DMF versus placebo was derived from the DEFINE[16] and CONFIRM[17] trials in RRMS. Using these trial data, four different scenarios were outlined

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TABLE 1. Results from the Prespecified Secondary Analysis Bayesian Model with Prior Probability According to Four Scenarios

	Prior Distribution Parameters		Likelihood Parameters	Posterior Distribution Parameters		Power (%)
Scenario 1 (early RRMS, subgroup of the trials)	Log (HR) = −1.45, SD = 0.31	HR = 0.24, 95% confidence interval (0.13–0.43)	Log (HR) = −2.66 SD = 0.97, HR = 0.07 95% confidence interval (0.01–0.45)	Log (HR) = −1.61 SD = 0.29	HR = 0.20, 95% credible interval (0.11–0.35)	90
Scenario 2 (all RRMS in the trials)	Log (HR) = −0.548, SD = 0.07	HR = 0.58, 95% confidence interval (0.50–0.66)		Log (HR) = −0.57 SD = 0.07	HR = 0.57, 95% credible interval (0.49–0.65)	100
Scenario 3 (all RRMS with an effect diluted by 20%)	Log (HR) = −0.362, SD = 0.07	HR = 0.70, 95% confidence interval (0.60–0.79)		Log (HR) = −0.38 SD = 0.07	HR = 0.68, 95% credible interval (0.59–0.78)	100
Scenario 4 (all RRMS with an effect diluted by 30%)	Log (HR) = −0.282, SD = 0.07	HR = 0.75, 95% confidence interval (0.65–0.86)		Log (HR) = −0.31 SD = 0.07	HR = 0.73, 95% credible interval (0.64–0.84)	100

Abbreviations: HR = hazard ratio; RRMS = relapsing–remitting multiple sclerosis; SD = standard deviation.

(Table 1). The scenario used for the analysis was based on the treatment effect in the subgroup of people with "early MS", defined as those with no previous treatment, less than 240 weeks since clinical disease onset and EDSS ≤ 1.5 (scenario 1 of 4; Table 1). The prior distribution had a hazard ratio (HR) = 0.23 (95% confidence interval (CI) = 0.13–0.43). Using this Bayesian approach, 40 participants per arm were needed to have 90% power to detect a 50% treatment effect.

An intention-to-treat analysis was performed on all randomly assigned participants. Both adjusted and unadjusted Cox proportional-hazards regression models were used to assess the impact of treatment from the time from randomization to the first acute or initial progressive neurological event (pre-specified primary endpoint). The adjusted model used key baseline variables for potential confounding effects. These variables included sex, age at the time of RIS diagnosis, MS family history, EDSS, brain T2-weighted hyperintense lesion volume (log-transformed) and presence of gadolinium-enhancing lesions. Survival curves were computed according to the Kaplan–Meier method to display the treatment effect.

The treatment effect on the time to the first clinical symptoms was estimated using HRs and 95% confidence intervals.

Analyses of MRI outcomes included participants who had at least one follow-up MRI. The cumulative mean number of new and/or newly-enlarging lesions on T2-weighted MRI and the cumulative mean number of new gadolinium-enhancing lesions over 96 weeks were compared between the two study groups by a negative binomial regression model. Adjusted analyses included the number of gadolinium-enhancing lesions on MRI at baseline as the covariate.

The change in mean lesion volume on T2-weighted MRI from baseline to week 96 was compared between the two study groups using a linear mixed model, which included a random intercept to account for the within-participant correlation. The T2-lesion volume was log-transformed for the analysis.

An independent external data safety monitoring committee reviewed all reported adverse events and continually monitored risk–benefit data.

All statistical analyses were performed using SAS (version 9.4) and R (version 4.1.3) The study is registered at ClinicalTrials.gov, NCT02739542 (ARISE).

Results

From March 9, 2016, to October 31, 2019, a total of 87 people with RIS were enrolled from 12 centers (Fig 1). The baseline demographic, clinical data, and reasons for the index MRI study revealing incidental anomalies highly

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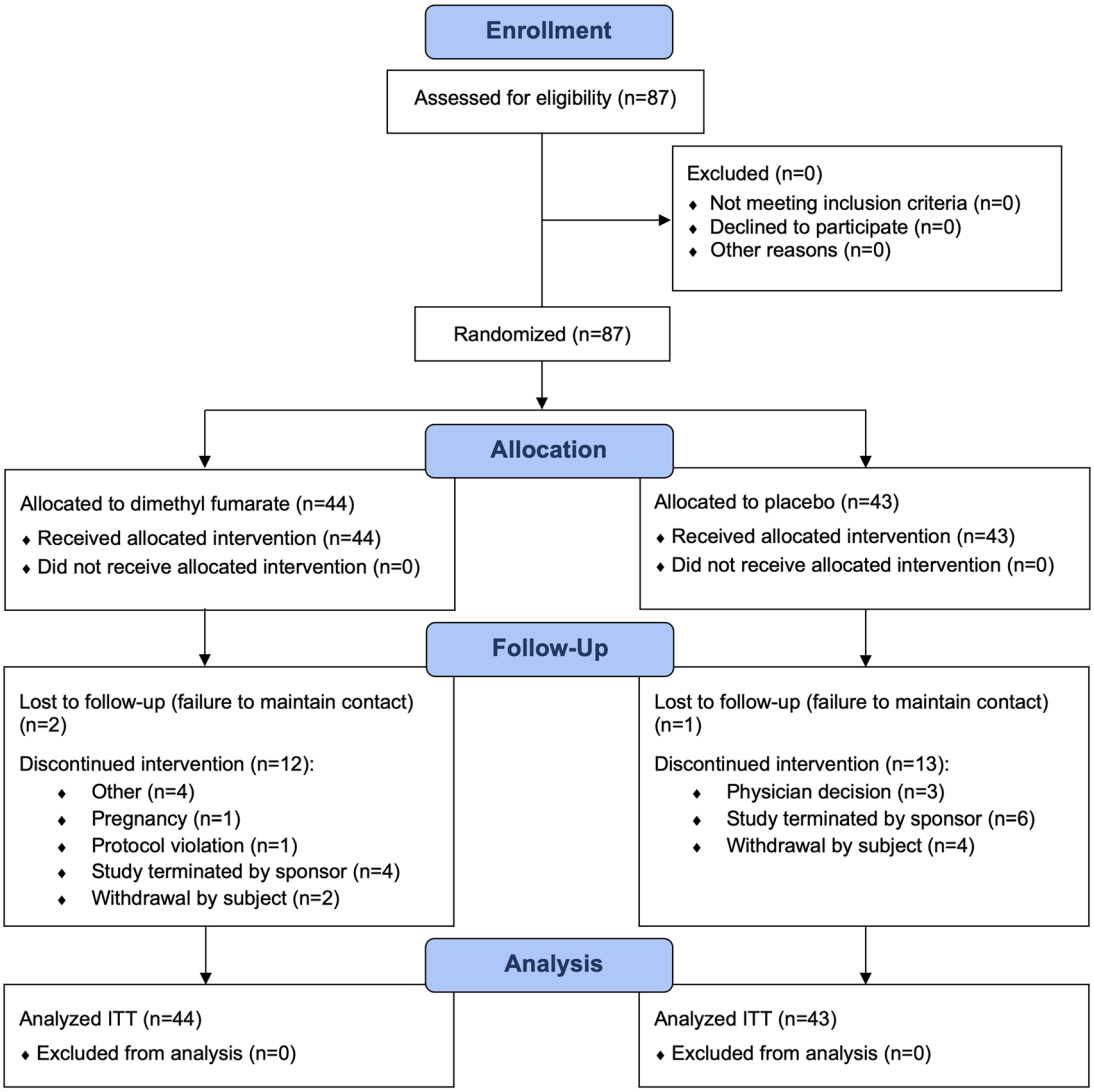

FIGURE 1: CONSORT flow diagram.

suggestive of CNS demyelination were generally well-balanced between groups apart from greater family history for MS in the placebo group and a higher frequency of gadolinium-enhancing lesions observed at baseline in the DMF group (Table 2, Supplementary Table 1). Median EDSS scores were 1.0 in both groups due to blinded examinations capturing physical impairments resulting from trauma, recent surgical procedures, and other well-documented non-CNS-based existing medical conditions. The discontinuation rate was 14/44 (32%) in the DMF group and 14/43 (33%) in the placebo group, including 10/87 subjects that were unable to complete the study due to early termination by the sponsor related to pace of enrollment on March 31, 2021 (Fig 1). The total number of individuals unblinded due to an adverse event was 19 (8 were randomized to DMF and 11 to placebo). The frequency of moderate adverse events was higher in the DMF group as compared to placebo (34 vs 19, respectively), with most common reasons involving infections and musculoskeletal connective tissue disorders. Musculoskeletal connective tissue disorders also represented the most common adverse events within the placebo group. In general, a higher frequency of infections and reduced lymphocyte counts were identified between groups. Differences in the frequency of severe adverse reactions were not observed with three identified events in the DMF group and four in the placebo group. Invasive ductal breast carcinoma and prostate cancer in two unique individuals were observed in the DMF group. Additional details regarding adverse reactions experienced by participants are provided in Supplementary Table 2. A total of 59 (68%) participants completed the 96-week study. Follow-up MRI studies performed at 96 weeks were acquired in 53/87 (61%) of people, 26/43 (60%) within the placebo group and 27/44 (61%) in the DMF group.

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The proportion of study participants with a first acute neurological symptom associated with CNS demyelination after 96 weeks from study randomization was 33% in the placebo and 7% in the DMF groups. Progressive clinical courses were not identified. Of the three exacerbations identified in the DMF group, the first presenting symptoms localized to the brainstem (1) and spinal cord (2). Within the placebo group, a total of 14 seminal clinical events were identified with symptoms localizing to the brain (1), brainstem (1), spinal cord (8), long sensory (9) and motor tracks (3). Compared to placebo, the HR for the risk of a first clinical event during the 96 weeks following treatment with DMF was 0.18 in the unadjusted model, (95% CI = 0.05–0.63, $p = 0.007$) (Fig 2 and Table 3). This result corresponds to an 82% reduction in the risk for a first neurological symptom associated with an acute or progressive demyelinating event in participants with RIS.

Results from the adjusted Cox proportional-hazards regression model were consistent with a significant effect of DMF on reducing the risk of the first relapse, even after adjusting for sex, age at the time of diagnosis, MS family history, EDSS, T2-weighted hyperintense volume (log-transformed) and the presence of gadolinium-enhancing lesions (HR = 0.07, 95% CI = 0.01–0.45, $p = 0.005$; Table 3).

When using the Bayesian approach, compared to placebo, the HR for the risk of a first clinical event during the 96-week period following treatment with DMF was 0.20 (95% credible interval, 0.11, 0.35) (Table 4). These results correspond to an 80% reduction in the risk for a first neurological symptom associated with an acute demyelinating event in participants with RIS. The 95% credible interval indicates that the probability that DMF has no effect on the time to a first neurological event compared to placebo is <0.0001. All sensitivity analyses, including different scenarios, for the prior distribution yielded similar results (Table 1).

The proportion of people who experienced at least one new and/or newly-enlarging T2-weighted

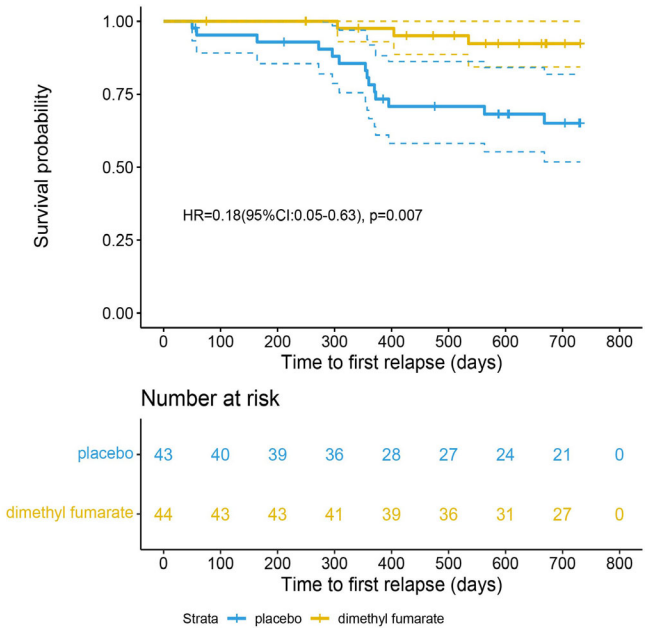

FIGURE 2: Results from the prespecified primary analysis utilizing Kaplan–Meier estimates of time from randomization to the first demyelinating event (unadjusted).

TABLE 2. Baseline Characteristics of the RIS Study Population

	Dimethyl Fumarate	Placebo
N	44	43
Age at RIS diagnosis (years), mean (SD)	43.57 (13)	44.81 (14)
Age at study entry (years), mean (SD)	43.73 (12)	46.25 (14)
Sex, n (%)		
Female	31 (70)	30 (70)
Male	13 (30)	13 (30)
Race, n (%)		
White	40 (91)	38 (88)
Black or African American	2 (5)	3 (7)
Other	2 (5)	2 (5)
Expanded disability status scale score, median (range)	1 (0–3)	1 (0–3)
MS family history, n (%)	3 (7)	6 (14)
Prior exposure to disease-modifying therapy (%)		
Interferon beta-1 alpha (Avonex®)	1 (2)	2 (5)
Interferon beta-1 beta (Betaseron®)	1 (3)	1 (2)
Glatiramer acetate (Copaxone®)	4 (9)	0 (0)
Presence of gadolinium enhancing lesions, n (%)	6/38 (16)	2/38 (5)
T2-weighted hyperintense lesion volume (log-transformed), mean (SD)	7.30 (1.07) N = 40	7.32 (0.89) N = 43

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TABLE 3. Effect of DMF on the Time to First Relapse According to the Prespecified Primary Analysis of Univariable (Unadjusted) and Multivariable (Adjusted) Cox Regression Models

	Unadjusted Analysis		Adjusted Analysis[a]	
	HR (95% CI)	*p*	HR (95% CI)	*p*
Treatment (dimethyl fumarate vs placebo)	0.18 (0.05–0.63)	0.007	0.07 (0.01–0.45)	0.005
Age at RIS diagnosis	0.93 (0.89–0.98)	0.003	0.92 (0.87–0.97)	0.004
Baseline expanded disability status scale (EDSS) score	0.70 (0.41–1.18)	0.180	0.99 (0.52–1.86)	0.967
Multiple sclerosis (MS) family history	0.54 (0.07–4.07)	0.550	0.31 (0.03–2.87)	0.304
Sex (male vs female)	1.63 (0.62–4.28)	0.322	2.70 (0.78–9.34)	0.118
T2-weighted hyperintense lesion volume (log-transformed) at baseline	0.76 (0.46–1.25)	0.276	0.85 (0.45–1.60)	0.610
Presence of Gd+ lesions at baseline	1.35 (0.30–6.02)	0.696	2.41 (0.35–16.65)	0.373

Abbreviation: CI: confidence interval.

[a]Adjustment for sex, age at the time of RIS diagnosis, MS family history, baseline EDSS score, T2-weighted hyperintense lesion volume (log-transformed) and presence of gadolinium enhancing lesions (Gd+) at baseline.

TABLE 4. Efficacy of Dimethyl Fumarate on Time to First Clinical Demyelinating Event

	Prior HR	Parameters for the Prior	Likelihood	Posterior Probability	HR (95% Credible Interval)
Scenario 1 (early RRMS, subgroup of the trial)	0.235	mu = −1.45 SD = 0.31	0.07 (0.01–0.46) mu = −2.70 SD = 0.98	mu = −1.62 SD = 0.29	0.20 (0.11–0.35)[a]

[a]Results from the prespecified secondary analysis Bayesian model with prior probability of outcome assumed to follow the dimethyl fumarate treatment effects in early RRMS within the pivotal DEFINE[16] and CONFIRM[17] phase 3 studies.

hyperintense lesion during the 96 weeks from study randomization was 31% (8/26) in the placebo and 26% (7/27) in the DMF group. Of those with MRI advancement, 6 of 8 in the placebo group and one of 7 in the DMF group converted to a first clinical event. Supplementary Table 3 contains the treatment status for participants who experienced a first clinical event, meeting the primary endpoint, at the time of the 96-week MRI study.

After adjusting for the number of gadolinium-enhancing lesions at baseline, DMF-treated subjects demonstrated significantly reduced cumulative number of new and/or newly-enlarging T2-weighted hyperintense lesions compared to placebo (0.12 vs 0.62, rate ratio = 0.20, 95% CI = 0.04–0.94, $p = 0.042$). Exposure to DMF was associated with a lower change in mean T2-weighted lesion volume over time in both the unadjusted and adjusted analyses; however, statistical significance was not achieved (0.005 cubic centimeters (cc) for DMF vs 0.04 cc for placebo, $p = 0.332$) (Table 5). In addition, as only one person within the DMF group had gadolinium-enhancing lesions at the 96-week assessment, statistical analysis of a treatment effect could not be performed. None of the participants within the placebo group demonstrated gadolinium-enhancement at 96 weeks.

At baseline, 37/87 (42%) of participants had EDSS scores higher than one, while at the 96-week visit, 35/57 (61%) had scores higher than one.

Discussion

In this first randomized, placebo-controlled study aimed at preventing the first clinical MS event in a cohort of individuals with RIS, we identified a treatment benefit of DMF in extending the time to clinical conversion and in the reduction of new and/or newly-enlarging T2-weighted hyperintense lesions over 96 weeks. These findings build

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TABLE 5. Summary from the Prespecified Secondary Unadjusted and Adjusted MRI Outcome Models

	Unadjusted Analysis				Adjusted Analysis			
	Dimethyl Fumarate	Placebo			Dimethyl Fumarate	Placebo		
	Mean Value (SD)		Rate Ratio (95% CI)	*p*	Mean Value (SD)		Rate Ratio (95% CI)	*p*
New/newly-enlarging T2-weighted hyperintense lesions	0.35 (0.19)	0.53 (0.28)	0.67 (0.15 to 2.96)	0.598	0.12 (0.07)	0.62 (0.30)	0.20 (0.04 to 0.94)	0.042
New gadolinium-enhancing lesions at week 96	0.07 (0.38)	0 (0.00)	–	–	–	–	–	–
			Mean Difference (95% CI)	*p*			Mean Difference (95% CI)	*p*
Change in lesion volume on T2-weighted MRI (log-transformed)	0.01 (0.03)	0.02 (0.03)	−0.14 (−0.71 to 0.43)	0.614	0.005 (0.03)	0.04 (0.03)	−0.31 (−0.96 to 0.33)	0.332

Note: Mean differences (95% CI) and *p*-values for the change in T2-weighted lesion volume were calculated with the use of a linear mixed model (with number of gadolinium-enhancing lesions at baseline as the covariate).
Note: Rate ratios (95% CI) and *p*-values for new/newly-enlarging T2-weighted hyperintense lesions were calculated with the use of negative binomial regression model (with number of gadolinium-enhancing lesions at baseline as the covariate).
Note: Rate ratios (95% CI) and *p*-values for new gadolinium-enhancing lesions at week 96 were calculated with the use of a negative binomial regression model.

on previous investigations focused on the study of individuals with clinically isolated syndrome (CIS), an early phase of the demyelinating spectrum in which individuals have experienced a first acute demyelinating event with risk for the possible future development of MS, whereby treatment with disease-modifying therapies have demonstrated benefit in reducing rates of clinical conversion. Our results are plausible as the observed features on MRI suggests that the biology related to CNS demyelination has already started and underscore the value of an opportunity for early DMT in delaying or preventing a first neurological symptom associated with CNS demyelination.

Given the spectrum of demyelinating disease from RIS to relapsing or progressive phases of MS,[10–13] the reported treatment effect and acquired safety data demonstrated consistency with prior DMF studies.[16, 17] The observed reductions in acute clinical events and frequency of new and/or newly-enlarging brain lesions resulting from immune-mediated injury are consistent with the anti-inflammatory and immunomodulatory effects of DMF. We demonstrated that DMF significantly reduced the development of new and/or newly-enlarging T2-weighted hyperintense lesions. The magnitude of the treatment effect on the clinical endpoint appeared more robust when compared to the MRI endpoints. These findings should be interpreted in the context of our study design. Following the development of a first clinical event, participants who were randomized to placebo were offered treatment with DMF. For those randomized to the active treatment arm, peg-interferon beta-1a was presented as the subsequent treatment and participants had the option to continue treatment with DMF. These aspects, coupled with the fixed frequencies of the MRI studies occurring at baseline and 96 weeks may clarify the interpretation of suppression of the number of T2-weighted hyperintense lesions, volume change, and scarcity of gadolinium-enhancing lesions observed. Moreover, the small number of clinical events causes the treatment effect estimate on the clinical endpoint to be highly variable, and the 95% CI includes much lower relapse reductions that are in line with those expected from the observed effect on MRI lesions.[22] Clinical courses consistent with primary progressive MS were not observed in either group which may be a reflection of the short study duration.

Pre-clinical therapeutic intervention before symptom manifestation is a contemporary theme in medicine. Early

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prevention studies exist in rheumatoid arthritis,[23] pre-diabetes,[24] and people with pre-hypertension.[25, 26] These approaches aim to reduce the emergence of clinical disease that modifies risk for future morbidity and mortality. Within the spectrum of neurodegenerative disorders, reduction in the accumulation of amyloid-ß peptide has been a central focus in preventing cognitive decline related to Alzheimer's disease.[27, 28] As a single clinical exacerbation related to MS may be associated with incomplete recovery and permanent neurological disability,[8, 29] the pursuit of preventative treatment, or one that delays the onset of clinical disease, with an established safety profile within the asymptomatic phase represents a critical solution.

Subjects with RIS have been well-characterized within the scientific literature. Risk for the clinical evolution to symptomatic states from large retrospective multinational cohorts include younger age, sex, presence of spinal cord and/or infratentorial lesions, CSF-restricted oligoclonal bands, and new T2-hyperintense lesion development on follow-up MRI studies.[9–11] An increase in the number of risk factors is associated with higher rates of clinical conversion. Within retrospective investigations, a significant number of people with RIS were found to be treated with a DMT with exposed subjects having higher rates of clinical conversion over time.[10] This earlier observation may have been related to treatment bias based on more concerning clinical and radiological profiles identified by the treating physician. Today, RIS subjects continue to be treated worldwide despite the lack of available data supporting benefit. Before this study, data demonstrating the impact of treatment from a randomized controlled trial within this phase of the MS spectrum were unavailable.

The application of our findings to the broader community should be addressed. At times, the accurate classification of RIS subjects has been challenged by the high sensitivity of MRI as non-specific white matter anomalies resulting from chronic migraine, microvascular disease, normal ageing, etc., may be erroneously characterized as originating from CNS demyelination.[30] The proper classification of subjects may also be hindered by reduced memory of medical information by individuals, a limitation that also extends to those with CIS and MS. All participants within this study were identified within tertiary centers specializing in neuroimmunology where clinical and radiological expertise may facilitate a more precise diagnosis of RIS. Hence, our results may not be entirely suitable for all general neurology practices where non-specific white matter anomalies may lead to the misdiagnosis of RIS resulting in psychological harm, avoidable medical procedures, and unnecessary DMT exposure.

The presented data should be evaluated in the context of several limitations. As this study was restricted to the United States, the impact of early treatment in RIS and generalizability of our findings in other geographical regions and populations worldwide is unknown. In addition, the study cohort was comprised primarily of white individuals. Therefore, the treatment effect on Black or African Americans, Asians, Hispanic or LatinX populations, and other non-white groups remains unclear. Similar to other randomized trials in MS, a determination of the presence of spinal cord lesions, a known risk factor in the clinical conversion of RIS subjects, was also not made. In addition, the withdrawal rate was identified to be over 30% between groups, which was significantly impacted by early study termination by the sponsor. Also, the treatment effect is very large, with wide confidence intervals. Therefore, further confirmation in larger studies is needed. Based on the type of clinical relapses identified in this study with a higher proportion of events localizing to the infratentorial region and spinal cord along with the observed event rates over 96 weeks, caution should be applied when considering uniform treatment recommendations given differences in likelihood for clinical conversion and risk for overtreatment. As our findings are based on population-based estimates, individualized decisions for care should be exercised, preferably by qualified experts in MS. Furthermore, expected outcomes following treatment may not also be identical to our reported results.

Our results show the effect of early treatment in preventing clinical conversion to MS, providing hope to people with RIS that clinical disease onset can be averted or delayed with early intervention when appropriately administered. Given fundamental differences between healthcare delivery systems and clinical practices, additional international efforts that aim to expand our findings are warranted. Also, the application of these findings when current dissemination in space definitions[31] are applied remain unknown. The future inclusion of CSF biomarkers and polygenic risk of MS[32] may also be of value in guiding the stratification of those with possible RIS and in estimating risk for a first clinical event. Longer-term follow-up studies are also needed to understand the impact of early treatment within this phase of the demyelinating disease spectrum on future disability outcomes. An increase in knowledge related to early treatment may not only result in modifications in how people are diagnosed and risk-stratified but also transform our approach to care toward effectively preventing or delaying the first onset of symptomatic disease itself.

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Author Contributions

D.T.O., O.K., C.L.-F., M.P.S., A.S., and D.P. contributed to the conception and design of the study; D.T.O., O.K., C.L.-F., M.P.S., C.J.A., F.B., L.H.H., L.A., E.M.M., W.S.V., S.D., R.B.T., A.O., G.P., R.P., O.S., A.S., and D.P. contributed to the acquisition and analysis of data; D.T.O., O.K., C.L.-F., M.P.S., A.S., and D.P. contributed to drafting the text or preparing the figures.

Potential Conflicts of Interest

Author D.T.O. has been a consultant to Biogen Inc., which owns patent rights to Tecfidera® that was used in this study. D.T.O. also received research support from Biogen Inc. and received royalties for intellectual property licensed by The Board of Regents of The University of Texas System. O.K. has been a consultant and also received research support from Biogen Inc. for an investigator-initiated study. C.L.-F. received research support from Biogen Inc. for an investigator-initiated study. M.P.S. has been a consultant and has received research grant support from Biogen Inc. L.H.H. received a research grant from Biogen Inc. E.M.M. received research support for an investigator-initiated study from Biogen Inc. C.H. holds stock in Biogen Inc; served as an employee of Biogen Inc. at the time of study concept, design, and execution. J.M. is a current employee at Biogen Inc. and holds stock. J.S.W. is a former employee of Biogen Inc. WSV has been a consultant to Biogen Inc. S.D. has been a consultant to Biogen Inc. R.T.N. has been a consultant to Biogen Inc. A.O. has been a consultant to Biogen Inc. G.P. has been a consultant to Biogen Inc. and has also received research grant support. P.R. has been a consultant to Biogen Inc. and has received research grant support. A.S. has been a consultant to Biogen Inc. C.J.A., F.B., L.A., C.v.H., O.S., and D.P. report no disclosures.

Data Availability Statement

Radiologically Isolated Syndrome Consortium (RISC) will share deidentified, individual participant data and other study information (study protocol, informed consent, statistical analysis plan, etc.) with academic investigators following publication and approval of a data access agreement by The University of Texas Southwestern Medical Center. All requests should be submitted to Darin T. Okuda, MD (darin.okuda@utsouthwestern.edu).

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